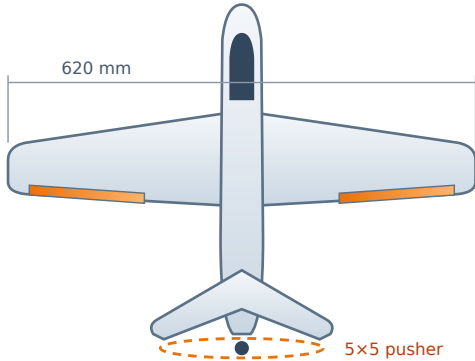
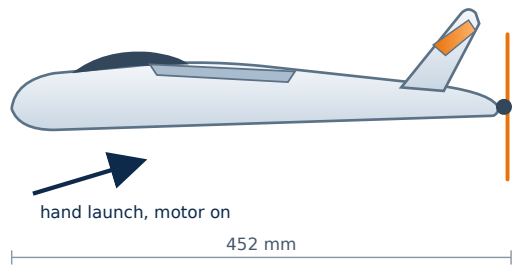


ZOHD Talon 250G

A 250-gram, hand-launched pusher V-tail — regulation-friendly weight, full-size capability.



TOP VIEW · span 620 mm



SIDE VIEW · length 452 mm

250 g

ALL-UP WEIGHT

6-16 m/s

SPEED ENVELOPE

4.3 m/s

MAX CLIMB

L/D 9

BEST GLIDE

T/W 1.0

STATIC THRUST

Airframe

Configuration	Pusher, V-tail, polyhedral wing
Wingspan	620 mm
Length	452 mm
Total lift surface	6.6 dm ² · AR 5.8
Wing loading	37.9 g/dm ²
Landing gear	None — belly lander
Controls	2 ailerons + 2 ruddervators (4× 4.3 g servos)

Powertrain

Motor	1406 brushless, 2600 KV
Propeller	5 × 5 in, 2-blade, tail pusher
ESC	30 A
Battery	2S Li-Ion · 8.4 V full / 6.0 V min
Static thrust	≈ 2.5 N (255 gf)

Performance (simulation-validated)

Max level speed	16 m/s (58 km/h)
Stall speed (1 g, power off)	6.7 m/s
Min speed, powered	≈ 6 m/s
Hands-off trim speed	9.5 m/s
Economic cruise	10-12 m/s @ 18-26 % throttle
Max climb rate	4.3 m/s ($\gamma \approx 24^\circ$)
Best glide	L/D ≈ 9 @ 10 m/s
Min sink	≈ 1.2 m/s
Roll rate (full aileron)	≈ 250 °/s
C _{L max} / stall α	1.50 / 16.5°

Why 250 grams? In many jurisdictions the 250 g class enjoys the lightest regulatory burden — this airframe delivers autopilot-grade stability, a 10 m/s speed spread and multi-hour endurance potential (battery dependent) while staying under the limit.

Hand launch, motor on

Belly landing — prop safe at tail

Paparazzi UAV / NPS ready

JSBSim flight-dynamics model

Flying the Talon 250G

Balance & Trim

Centre of gravity	114 mm aft of nose tip ($\approx 30\%$ MAC)
Hands-off trim	9.5 m/s at $\alpha \approx 6.5^\circ$
Power-on pitch	Mild pitch-up as speed builds — trim per throttle band
Lateral trim	Neutral; slow spiral if left hands-off > 30 s

Control Surface Throws

SURFACE	THROW	NOTES
Elevator	$\pm 20^\circ$	full throw reaches stall α
Ailerons	$\pm 20^\circ$	$\approx 250^\circ/\text{s}$; use 60 % low rates
Rudder	$\pm 25^\circ$	modest yaw, small roll couple
V-tail mixing	50 / 50 elevator-rudder mix in TX or autopilot	

Speeds to Remember

PHASE	IAS	THROTTLE
Never slower than (power off)	7.5 m/s	—
Approach	8-9 m/s	10-15 %
Economic cruise	10-12 m/s	18-26 %
Fast cruise	14 m/s	$\approx 45\%$
Maximum level	16 m/s	100 %

Battery Discipline (2S Li-Ion)

- Launch full: **8.4 V**. Li-Ion sags more than LiPo — softer climb late in flight.
- Plan landing at **6.6-6.4 V** under load; **hard floor 6.0 V**.
- Below 10 °C, warm the pack — cold Li-Ion sags hard on launch.

Hand Launch (motor ON)

- Full throttle **before** release — the 5×5 pusher is behind you, clear of your arm.
- Firm level throw, ≥ 9 m/s, nose **8-12°** up, wings level.
- Hands off is fine: the aircraft accelerates and settles into a $\approx 20^\circ$ climb on its own.
- Weak flat throws (≤ 8 m/s, $< 6^\circ$) will mush and settle** — commit to the toss.

Landing (belly)

- Approach 8-9 m/s, power near idle; the clean airframe floats — plan a long shallow final.
- Kill throttle over the threshold; flare to $\approx 10^\circ$ and let it settle on the belly.
- Prop is protected at the tail; grass or dirt strips preferred over gravel.

Handling Peculiarities

- Pusher throttle coupling** — power changes pitch trim noticeably; add throttle smoothly on approach.
- No prop wash over tails** — unlike a tractor, controls do NOT firm up with throttle at low airspeed; keep airspeed up during go-arounds.
- Stall** — gentle mush at 16.5° α with mild wing drop; recovery is immediate on relaxing elevator. Height loss $\approx 5-8$ m.
- Spiral mode** — slowly divergent ($\approx 20^\circ$ bank per minute hands-off); normal for the class, the autopilot holds it easily.
- Wind** — at 37.9 g/dm² loading, stay below ≈ 8 m/s wind; penetration reserve is 16 m/s V_{max} .
- Adverse yaw** — small; a touch of rudder mix (10-15 %) tidies up turn entries.

Paparazzi / NPS Simulation

- Airframe:
conf/airframes/OPENUAS/openuas_zohd_talon_250g.xml
- FDM: **JSBSIM_MODEL = OPENUAS/openuas_jsbsim_zohd_talon_250g**
- Motor response modeled with 0.1 s lag; thrust via external reaction (validated on JSBSim 1.1.14 & 1.3.1).

One-line character reference: docile at cruise, honest at the stall, launch it with conviction, manage pitch with throttle in mind, and respect the Li-Ion floor.